

Response submitted by Chile

1. Has your Government previously issued an official paper, speech, or similar statement summarizing how it understands international law applies to cyber operations? Please provide copies or links to any such statements.

Supreme Decree N° 3, whereby the Ministry of Defense established the National Cyber-Defense Policy, was published on March 9, 2018 in the Official Gazette of the Republic of Chile.

<https://www.diariooficialinterior.gob.cl/publicaciones/2018/03/09/42003/01/1363153.pdf>

2. Do existing fields of international law (including the prohibition on the use of force, the right of self-defense, international humanitarian law, and human rights) apply to cyberspace? Are there areas where the novelty of cyberspace excludes the application of a particular set of international legal rights or obligations?

The Republic of Chile has stated, through its National Cyber-Defense Policy, that it will refrain from resorting to the threat of the use or the use of force through cyberspace in a manner that is against international law, and that it will exercise its right to self-defense against any armed attack carried out through cyberspace, in accordance with the provisions of Article 51 of the United Nations Charter. The Republic of Chile has pointed out furthermore that the planning, conduct, and execution of operations in cyberspace must adhere strictly to respect for public international law, with particular consideration to international human rights law and international humanitarian law.

Consequently, Chile is of the view that current international law provides the applicable legal framework that should regulate how states behave in cyberspace, including rules relating to *jus ad bellum*, international humanitarian law, human rights, and those governing the international responsibility of states. Likewise, Chile believes that any additional rules that may emerge in the future to specifically regulate the conduct of states in cyberspace must be consistent with existing international law.

3. Can a cyber operation by itself constitute a use of force? Can it constitute an armed attack that triggers a right of self-defense under Article 51 of the UN Charter? Can a cyber operation qualify as a use of force or armed attack without causing the violent effects that have been used to mark such thresholds in past kinetic conflicts?

Article 2.4 of the United Nations Charter states that the Members of the Organization shall refrain in their international relations from the threat or use of force against the territorial integrity or political independence of any state, or in any other manner inconsistent with the purposes of the United Nations.

While the Charter establishes no definition of the use of force, the prohibition has traditionally been understood as referring exclusively to armed force, excluding other forms of political and economic coercion. Besides, the International Court of Justice has stated that both Article 2.4 and Article 51 – the latter enshrining the right to self-defense – are applicable for use of force, regardless of the weapons used, for example, through cyberattacks.

A cyber operation may therefore constitute a use of force *per se* in those cases in which its scale and effects are comparable to those of a cyber or non-cyber operation rising to the level of use of force. In other words, it is not the means used but rather the effects on a state as a result of such an operation, cyber or kinetic, that determine whether there is use of force.

In its National Cyber Defense Policy, the Republic of Chile has articulated its view that a cyberattack can be as harmful as an armed attack. Accordingly, the Republic of Chile is of the opinion that cyberattacks directed against its sovereignty, its inhabitants, its physical or information infrastructure, or cyberattacks which severely affect its interests and satisfy the threshold of severity necessary as well as the requirements established by international law to be an armed attack, will give rise, in accordance with Article 51 of the United Nations Charter, to the right to self-defense, which may be exercised through physical means as well through cyberspace.

4. Outside of armed conflicts, when would a State be responsible for the cyber operations of a non-State actor? What levels of control or involvement must a State have with respect to the non-State actor's operations to trigger the international legal responsibility of that State?

A state shall be held responsible for an internationally unlawful cyber operation that is carried out through any of its organs, by any person or body exercising governmental authority, or by a person or group of persons acting under instructions from, or at the direction or control of, that state.

In the latter case, the degree or level of control or involvement of a state in the operations of a non-state actor, required to trigger its international responsibility is that of effective control, as the International Court of Justice has ruled on two occasions (*Military and Paramilitary Activities in and against the Government of Nicaragua - Nicaragua v. United States, 1986*; and *Application of the Convention on the Prevention and Punishment of the Crime of Genocide - Bosnia and Herzegovina v. Serbia and Montenegro, 2007*).

Hence, with regard to cyber operations carried out by non-state actors, a state shall be held responsible for such conduct if it is confined to one of the scenarios outlined above.

5. Are the standards of State responsibility the same or different in the context of an armed conflict as that term is defined in Articles 2 and 3 common to the 1949 Geneva Conventions?

The normative framework concerning responsibility of a state establishes secondary rules applicable to all primary rules of general international law, including those applicable in the context of an armed conflict, as the term is defined in common Articles 2 and 3 of the 1949 Geneva Conventions.

6. Under international humanitarian law, can a cyber operation qualify as an “attack” for the rules governing the conduct of hostilities if it does not cause death, injury or direct physical harm to the targeted computer system or the infrastructure it supports? Could a cyber operation that produces only a loss of functionality, for example, qualify as an attack? If so, in which cases?

It should be noted *a priori* that cyber operations carried out by one state against another state could constitute an attack, under international humanitarian law.

Article 49 of the Protocol Additional to the Geneva Conventions on the Protection of Victims of International Armed Conflicts (Protocol I) states that “attacks” means acts of violence against the adversary, whether in offense or in defense.

The acts of violence referred to in Protocol I are not limited exclusively to those of a kinetic nature.

The key point in determining whether an attack exists is the effects or consequences arising from the act itself, which must be violent. Thus, under the rules governing the conduct of hostilities, a cyber-operation, whether in offense or in defense, could constitute an attack in cases wherein it is reasonably expected to result in injury, death of persons, or physical damage to property.

In the cyber arena, it is possible for an operation to have no material consequences but be aimed at disabling or ceasing the functionality of a cyber-system or infrastructure, affecting the capabilities of a state under attack. In this context, what is relevant is to determine whether, as a result of the attack, the affected state must take action to repair or restore the affected infrastructure or computer systems, since in those cases the consequences of the attack are similar to those described above, in particular physical damage to property.

7. Is a cyber operation that only targets data governed by the international humanitarian law obligation to direct attacks only against military objectives and not against civilian objects?

As a preliminary matter, it should be noted that a cyber-operation could only be regulated by international humanitarian law to the extent that it takes place in the context of an armed conflict.

Should this be the case, Article 52 of the Protocol Additional to the Geneva Conventions on the Protection of Victims of International Armed Conflicts (Protocol I) would be the pertinent rule in this matter. It states that:

1. Civilian objects shall not be the object of attack or of reprisals. Civilian objects are all objects that are not military objectives as defined in paragraph 2.
2. Attacks shall be limited strictly to military objectives. Insofar as objects are concerned, military objectives are limited to those objects which by their nature, location, purpose, or use make an effective contribution to military action and whose total or partial destruction, capture or neutralization, in the circumstances ruling at the time, offers a definite military advantage.

This distinction is fundamental to international humanitarian law and is considered an expression of international custom, in relation to international as well as non-international armed conflicts.

With respect to a cyber-operation that only targets computer data, the central point is to define whether the latter qualify as an object wherein a distinction can be made between a military and a civilian target. The Commentary on the Additional Protocols of June 8, 1977 to the 1949 Geneva Conventions, prepared by the International Committee of the Red Cross in 1987, states that objects are characterized as being “visible and tangible.” From this perspective, under current international humanitarian law the aforementioned data would not qualify as objects, in principle, because they are essentially intangible, without prejudice to the physical elements containing the data – hardware, for example.

Nevertheless, it should be borne in mind that an attack directed exclusively at computer data could very well produce adverse consequences affecting the civilian population, such as in the case of the elimination, through a cyber-operation, of a database of social security for the population.

The principle of distinction must therefore be taken into consideration in the context of cyber operations, whereby a state should refrain from attacking data in case it could affect the civilian population, unless such data are being used for military purposes.

8. Is sovereignty a discrete rule of international law that prohibits States from engaging in specific cyber operations? If so, does that prohibition cover cyber operations that fall below the use of force threshold and which do not otherwise violate the duty of non-intervention?

The State of Chile, through its National Cyber Defense Policy, has declared that it will protect its critical information infrastructure by exercising its sovereignty over those networks and digital resources.

Cyber operations occur in a particular territory and involve objects, or are carried out by persons or entities over which the state may exercise sovereignty. The State of Chile has sovereignty over a cyber or information infrastructure located in its territory and over activities associated with that infrastructure.

Accordingly, a state is free to take any step it considers necessary or appropriate with respect to its cyber infrastructure as well as in relation to cyber operations that have taken place on that infrastructure and with respect to individuals involved in them, as long as those measures do not violate the rule of international law – for example, those contained in international human rights law or international humanitarian law.

States carrying out cyber operations must always take the sovereignty of other states into account. Because cyberspace is interconnected, cyber operations carried out in one state can have effects on another sovereign state. Hence, every time a state considers carrying out a cyber-operation, it should consider ensuring it does not affect the sovereignty of another. It should also be added that in keeping with the National Cyber-Security Policy issued via Presidential Order N° 1 (2017), the guidelines of the State of Chile in the field of cyberspace are that it be free, open, secure, and resilient, which must undoubtedly be related to respect for sovereignty of another state in this arena.

Likewise, based on the principle of sovereignty, every state has an obligation to respect the territorial integrity and independence of other states and must faithfully discharge its international obligations, including as regards the principle of nonintervention. Cyber operations that hinder another state from exercising its sovereignty therefore constitute a violation of that sovereignty and are prohibited under international law.

9. Does due diligence qualify as a rule of international law that States must follow in exercising sovereignty over the information and communication technologies in their territory or under the control of their nationals?

Due diligence is a general principle of international law, derived from sovereignty of states. Accordingly, not only does the principle of sovereignty guarantee rights to states but it also imposes legal obligations, including the obligation to prevent their territory, and those objects over which they exercise sovereignty, from being used to cause harm to other states. This principle has long been recognized in the jurisprudence of the International Court of Justice, since the April 9, 1949 judgment in the Corfu Channel case, in which the court stated:

“it is every State’s obligation not to allow knowingly its territory to be used for acts contrary to the rights of other States”

The due diligence obligation applies to a state’s entire sovereign territory, including all cyber infrastructure, as well as persons carrying out cyber operations within its territory.

Therefore, from a cyber-operations standpoint, a state must exercise due diligence to prevent its sovereign territory, including the cyber infrastructure under its control, from being used to carry out cyber operations that affect another state's rights or could have adverse consequences for them.

Application of the principle of due diligence, as a binding rule with respect to information and communication technologies in their territories, has been recognized by the United Nations Group of Governmental Experts on Developments in the Field of Information and Telecommunications in the Context of International Security. The Group of Experts has specifically stated that the international law principles derived from state sovereignty are binding in the context of cyber operations.

10. Are there other rules of international law that your government believes are important to highlight in assessing the regulation of cyber operations by States or actors for which a State is internationally responsible?